CHLOE HALLAERT

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TECHNICAL SKILLS

Languages	Python, C++, C, SQL, Bash, TypeScript/JavaScript, Go (basic)
Backend & Systems	APIs, gRPC/Protobuf, multithreading, concurrency, Linux, ROS 2, FreeRTOS, embedded systems
DevOps & Cloud	Docker, Git, CI/CD (GitHub Actions), pytest, unit testing, AWS (EC2), GCP, Azure, SQLite, Milvus
ML & Tools	PyTorch, TensorFlow, TensorRT, ONNX, MLflow, NumPy, Pandas, OpenCV, ffmpeg, Sphinx, addr2line

EDUCATION

Bachelor of Software Engineering, Honors — University of Waterloo **2024-Present**

- Coursework: Data Structures & Algorithms, Software Design (C++), Databases, Compilers, Computer Architecture
- Awarded International Student Scholarship (one of 20 recipients) and President's Scholarship

EXPERIENCE

Robotics Software Engineer |*Python, C++, C, gRPC/Protobuf, ROS 2, Docker, GitHub Actions, Linux* Sept 2026 - Present
Mara Inc *San Francisco, California, United States*

- Built a **gRPC/Protobuf** code-generation pipeline (**buf**) producing **Python, TypeScript, Go and C++ SDKs**, with **CI/CD** (GitHub Actions, Docker) for build checks and **versioned release artifacts**
- Root-caused a silent **HardFault** (status still published, but every command failed): ruled out 3 hypotheses, preserved the fault address across resets, and decoded it with `addr2line` to a **dangling pointer** to stack memory
- Cut **1.3 s** of latency from a **multithreaded**, **unit-tested** ROS 2 video node by replacing a buffering decoder with **ffmpeg**
- Built **real-time embedded firmware** (**C**, **FreeRTOS**, STM32) replacing an off-the-shelf motor controller, with a **ROS 2** action server; moved step generation to a **hardware timer (PWM)**, removing jitter and cutting actuation time to ∼**0.4 s**

AI Engineer |*Python, PyTorch, NumPy, SQLite, MLflow, Sphinx, Docker* Jan - May 2026
Mara Inc *San Francisco, California, United States*

- Built a **Python CLI** that cut dataset preparation time by **97%**, saving an estimated **$100K**, using **statistical analysis** of telemetry to detect usable segments, **linear interpolation**, and coordinate transforms to auto-generate labels
- **Refactored** the tool into a modular **MVC** architecture after **code review**; auto-generated **Sphinx** docs from docstrings
- Improved a **transformer**-based tracking model's **success rate (AUC)** by **350%** by designing a **custom loss function** (asymmetric loss, CIoU with size penalties, label smoothing) and drift-guarded template updates; shipped to **production**
- Delivered an autonomous system **field deployment** in **3 days** for a high-stakes investor demo, owning software setup, hardware bring-up and live debugging

AI & Software Engineer |*Python, AWS, Docker, Milvus, Selenium, BS4, SQLite, SQL, Transformers* Sept 2024 - July 2025
Wilson & Wilbur Consulting Group *Toronto, Ontario, Canada*

- Integrated the **LLaMA 1B** LLM into a **Dockerized chat service**, exposing **API endpoints** for conversation, summarization, and reasoning; designed prompts, evaluated outputs with **Transformers/evaluate**, and implemented **unit tests**
- Developed a Retrieval-Augmented Generation **(RAG) pipeline** and deployed a vector database service using **Milvus** and **Docker**, implementing **CRUD operations** on vector databases to improve efficiency and reduce upload time by **70%**
- Built and deployed a **web scraper** using **Selenium** and **Beautiful Soup**, leveraging **multithreading** and **multiprocessing** to scale automated data ingestion, implemented unit tests, deployed to **AWS EC2**, and optimized data extraction with **SQL/SQLite**

PROJECTS

Object Detection & Semantic Navigation System |*Python, PyTorch, TensorFlow, ROS, YOLOv5, LIDAR, pandas*

- Developed an object detection and semantic navigation system using **Python**, **PyTorch**, **TensorFlow**, and **ROS** for real-time perception, tracking, and probabilistic semantic mapping for ground robot navigation **[demo]**
- Built a **custom neural network** and integrated **YOLOv5** and a custom binary classification model for real-time detection and adaptive navigation; fused sensor and visual input data using **pandas** for streamlined data manipulation

ADDITIONAL

NASA Human Exploration Rover Challenge – Team lead of 8; led design, scheduling and safety review of a Mars rover
Brown University Pre-College – Mathematical Modeling of Finance: Intro to Quantitative Analysis
IB Bilingual Diploma | **Published Author** of two children's educational books